

Yash Agrawal

AI/ML Engineer

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PROFILE

AI/ML Engineer with experience in Large Language Models (LLMs), prompt engineering, LLM fine-tuning, AI evaluation, data annotation, and industrial automation. Skilled in Python, SQL, NLP, RAG, and PLC programming with hands-on experience developing AI-powered automation solutions.

EXPERIENCE

AI/LLM Analyst – Innodata Inc.

Jan 2026 – Present

- Designed and optimized prompts for Large Language Models (LLMs) to improve response quality and instruction adherence.
- Performed data annotation, validation, and quality assurance for AI model training and evaluation.
- Evaluated AI-generated responses for factual accuracy, relevance, safety, and completeness.
- Conducted prompt engineering, re-prompting, and comparative response analysis to improve model performance.

R&D Intern – Schneider Electric

Mar 2025 – Aug 2025

- Developed industrial automation applications using IEC 61131-3 and IEC 61499 standards.
- Designed and developed an AI-powered pipeline to automate IEC 61131-3 to IEC 61499 conversion.
- Built and curated domain-specific datasets from Function Block Diagram (FBD) and Structured Text (ST) programs.
- Fine-tuned a Large Language Model (LLM) using the curated datasets for automated PLC code conversion.
- Parsed PLC XML project files, extracted function blocks, variables, and connections, and generated IEC 61499-compliant XML using Python.
- Evaluated, validated, and refined model outputs through iterative prompt engineering and testing.
- Proposed a Retrieval-Augmented Generation (RAG) pipeline leveraging domain-specific documentation to reduce hallucinations and improve conversion accuracy.

EDUCATION

Bachelor of Technology (B.Tech.) in Electronics and Instrumentation Engineering

Manipal Institute of Technology, Manipal | 2021 – 2025

PROJECTS

AI-Powered IEC 61131-3 to IEC 61499 Converter

- Developed an AI-powered pipeline to automate conversion of IEC 61131-3 PLC applications into IEC 61499 function blocks.
- Built and curated domain-specific datasets from FBD and Structured Text (ST) programs.
- Parsed PLC XML files and generated IEC 61499-compliant XML using Python.
- Fine-tuned an LLM and improved output quality through prompt engineering, validation, and testing.

- Proposed a RAG pipeline using IEC standards and software documentation to improve conversion accuracy.

Smart IoT Parking System with ANPR

- Developed an IoT-based smart parking system using ESP32, ESP32-CAM, MQTT, and Node-RED.
- Implemented Automatic Number Plate Recognition (ANPR) for vehicle authentication.
- Integrated LDR, IR sensors, and servo motors for occupancy detection and automated gate control.
- Designed a real-time Node-RED dashboard for parking monitoring.

Dental Caries Detection using MATLAB

- Developed an image-processing algorithm to detect dental caries from X-ray and intraoral images.
- Applied preprocessing, segmentation, and feature extraction techniques to improve detection accuracy.

SKILLS

Python • SQL • MATLAB • Prompt Engineering • LLM Fine-tuning • Data Annotation • AI Evaluation • RAG • NLP
• XML Parsing • PLC Programming • IEC 61131-3 • IEC 61499 • Node-RED • MQTT • Git

CERTIFICATIONS

- Data I/O and Preprocessing with Python and SQL – DeepLearning.AI
- Anthropic – Introduction to MCP
- Google Cloud Digital Leader Training